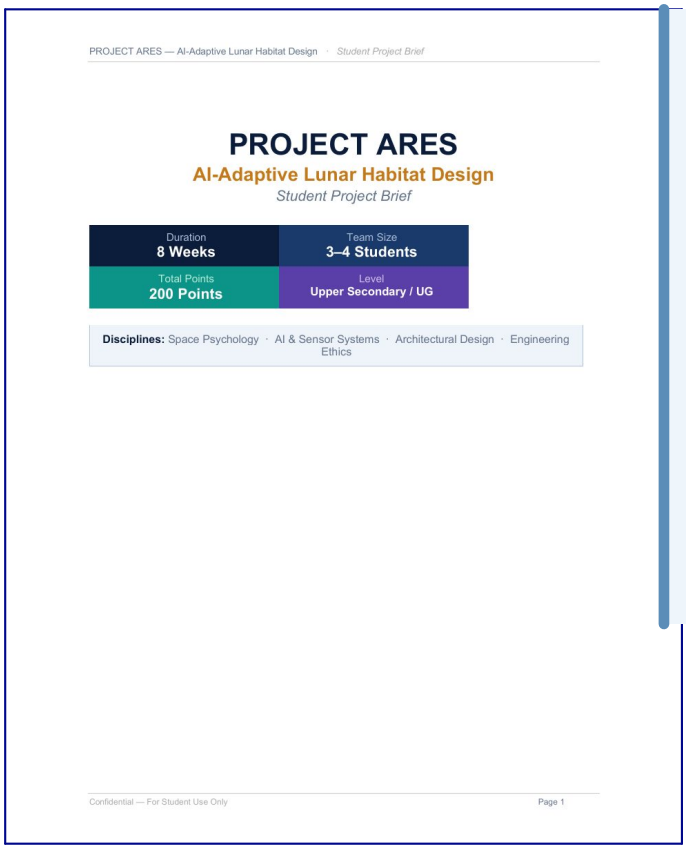
Project ARES Student Brief (with our notes)

This is our copy of the Project ARES Student Brief. Under each page we wrote what we did for it, how we did it, and which part of the rubric it covers, with screenshots from both of our games: the **realistic game** (Moon_Base_Final) and the **sci-fi game** (Lunar Habitat Commander).

Done means it's fully in the game. **Mostly done** means it's in there but something is still missing, and we say what.

We graded ourselves pretty strictly and got **189 out of 220**. The full table is on page 17.


PROJECT ARES — AI-Adaptive Lunar Habitat Design · Student Project Brief

PROJECT ARES

AI-Adaptive Lunar Habitat Design

Student Project Brief

Duration 8 Weeks | Team Size 3–4 Students

Total Points 200 Points | Level Upper Secondary / UG

Disciplines: Space Psychology · AI & Sensor Systems · Architectural Design · Engineering Ethics

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Done | Cover page

- We made two full simulators instead of one. The realistic game is a small habitat with 4 crew, and the sci-fi game (Lunar Habitat Commander) is a bigger base with an AI helper bot and portals. Both cover the whole brief.
- Both work in VR on a Meta Quest and on a normal PC.
- When you start either game, guided missions walk you through everything. In the realistic game you just keep pressing ENTER, and in the sci-fi game it's G. A judge can see the whole project in a few minutes this way.

How: We built both in Unity with C#. The AI part (the six conditions, nine sensors and three tiers) actually runs inside the games, it's not just slides.

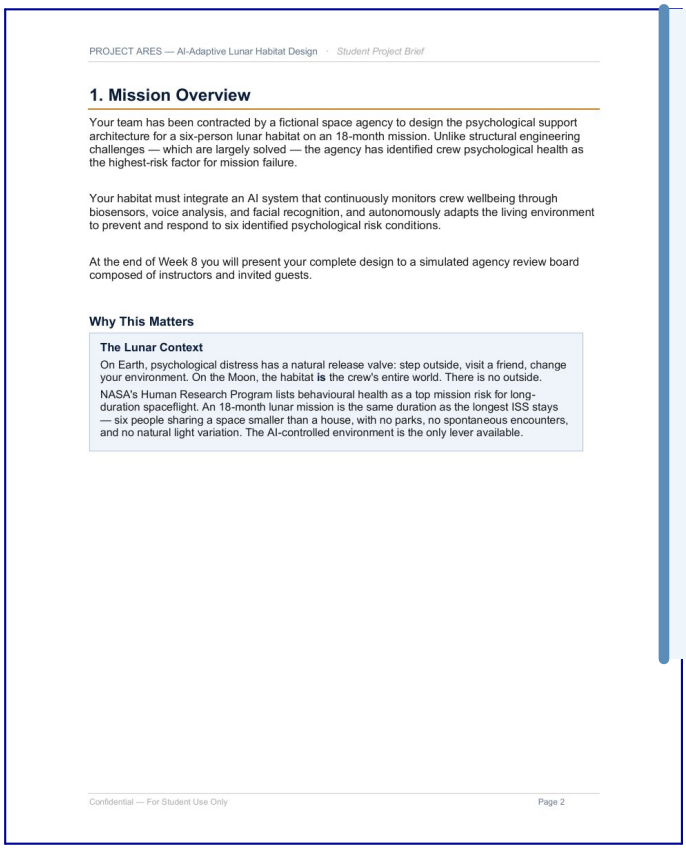
Screenshots:



Realistic game, first mission



Sci-fi game, mission card



PROJECT ARES — AI-Adaptive Lunar Habitat Design · *Student Project Brief*

1. Mission Overview

Your team has been contracted by a fictional space agency to design the psychological support architecture for a six-person lunar habitat on an 18-month mission. Unlike structural engineering challenges — which are largely solved — the agency has identified crew psychological health as the highest-risk factor for mission failure.

Your habitat must integrate an AI system that continuously monitors crew wellbeing through biosensors, voice analysis, and facial recognition, and autonomously adapts the living environment to prevent and respond to six identified psychological risk conditions.

At the end of Week 8 you will present your complete design to a simulated agency review board composed of instructors and invited guests.

Why This Matters

The Lunar Context

On Earth, psychological distress has a natural release valve: step outside, visit a friend, change your environment. On the Moon, the habitat **is** the crew's entire world. There is no outside.

NASA's Human Research Program lists behavioural health as a top mission risk for long-duration spaceflight. An 18-month lunar mission is the same duration as the longest ISS stays — six people sharing a space smaller than a house, with no parks, no spontaneous encounters, and no natural light variation. The AI-controlled environment is the only lever available.

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Mostly done | Mission overview

- The AI watches the crew through wearables, plus voice and face analysis in shared rooms only. It then changes the light, sound, temperature, scent, air and schedule on its own.
- We treated the habitat as a mental health design, not just a building. Every room has a job for the crew's wellbeing (see pages 7 and 8).

How: Each crew member has their own stats in the game (stress, mood, tiredness, sleep, cortisol and so on). The sensors turn those into readings, and the AI turns the readings into the six condition scores every few seconds.

Still need to do: The brief says 6 crew but we used 4 (you plus 3). The NASA HUNCH page says 2 to 4 astronauts, so we went with that. We'll explain this to the board.

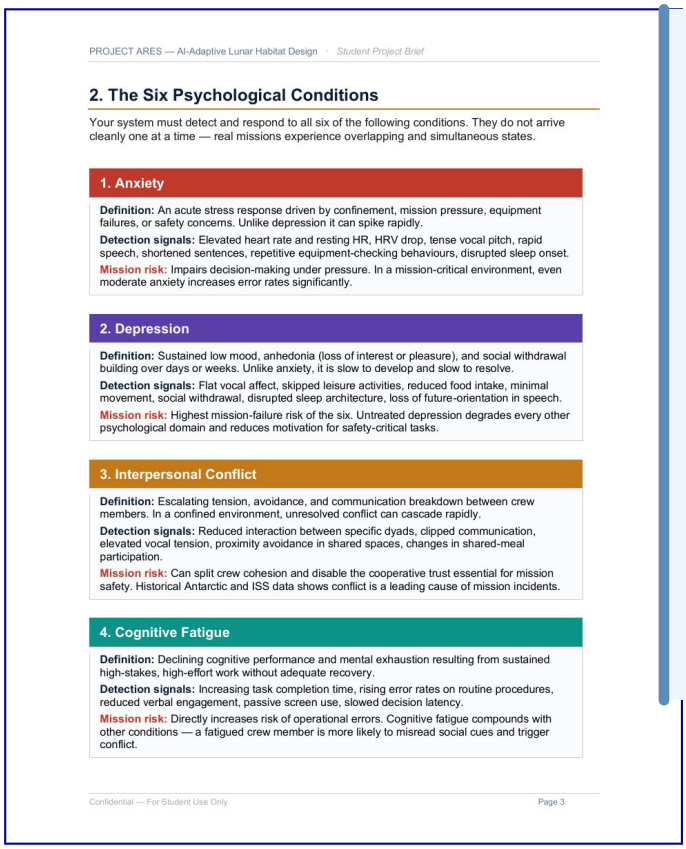
Screenshots:



Realistic game, checking on a crewmate in person



Sci-fi game, the same check-in on Dr. Okafor



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2. The Six Psychological Conditions

Your system must detect and respond to all six of the following conditions. They do not arrive cleanly one at a time — real missions experience overlapping and simultaneous states.

1. Anxiety

Definition: An acute stress response driven by confinement, mission pressure, equipment failures, or safety concerns. Unlike depression it can spike rapidly.

Detection signals: Elevated heart rate and resting HR, HRV drop, tense vocal pitch, rapid speech, shortened sentences, repetitive equipment-checking behaviours, disrupted sleep onset.

Mission risk: Impairs decision-making under pressure. In a mission-critical environment, even moderate anxiety increases error rates significantly.

2. Depression

Definition: Sustained low mood, anhedonia (loss of interest or pleasure), and social withdrawal building over days or weeks. Unlike anxiety, it is slow to develop and slow to resolve.

Detection signals: Flat vocal affect, skipped leisure activities, reduced food intake, minimal movement, social withdrawal, disrupted sleep architecture, loss of future-orientation in speech.

Mission risk: Highest mission-failure risk of the six. Untreated depression degrades every other psychological domain and reduces motivation for safety-critical tasks.

3. Interpersonal Conflict

Definition: Escalating tension, avoidance, and communication breakdown between crew members. In a confined environment, unresolved conflict can cascade rapidly.

Detection signals: Reduced interaction between specific dyads, clipped communication, elevated vocal tension, proximity avoidance in shared spaces, changes in shared-meal participation.

Mission risk: Can split crew cohesion and disable the cooperative trust essential for mission safety. Historical Antarctic and ISS data shows conflict is a leading cause of mission incidents.

4. Cognitive Fatigue

Definition: Declining cognitive performance and mental exhaustion resulting from sustained high-stakes, high-effort work without adequate recovery.

Detection signals: Increasing task completion time, rising error rates on routine procedures, reduced verbal engagement, passive screen use, slowed decision latency.

Mission risk: Directly increases risk of operational errors. Cognitive fatigue compounds with other conditions — a fatigued crew member is more likely to misread social cues and trigger conflict.

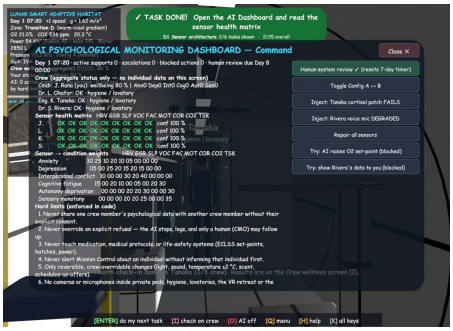
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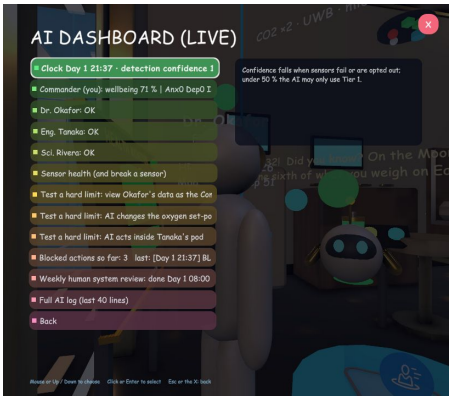
Done | The six conditions (anxiety, depression, conflict, fatigue)

- All six conditions are tracked for every crew member at the same time, so overlapping problems work too (that's board question Q2).
- Each condition has its own timing like the brief says. Anxiety reacts within minutes and gets watched for 20 minutes. Depression needs about 48 hours of signs and gets watched for 72 hours. Conflict and fatigue have their own times too.
- We used the warning signs from the brief: HRV dropping, a tense voice, bad sleep, flat expressions, people pulling away from each other, more mistakes and slower work.

How: The game has a table that says how much each sensor counts toward each condition. Every condition uses 4 to 6 sensors, so one bad reading can't set it off by itself.

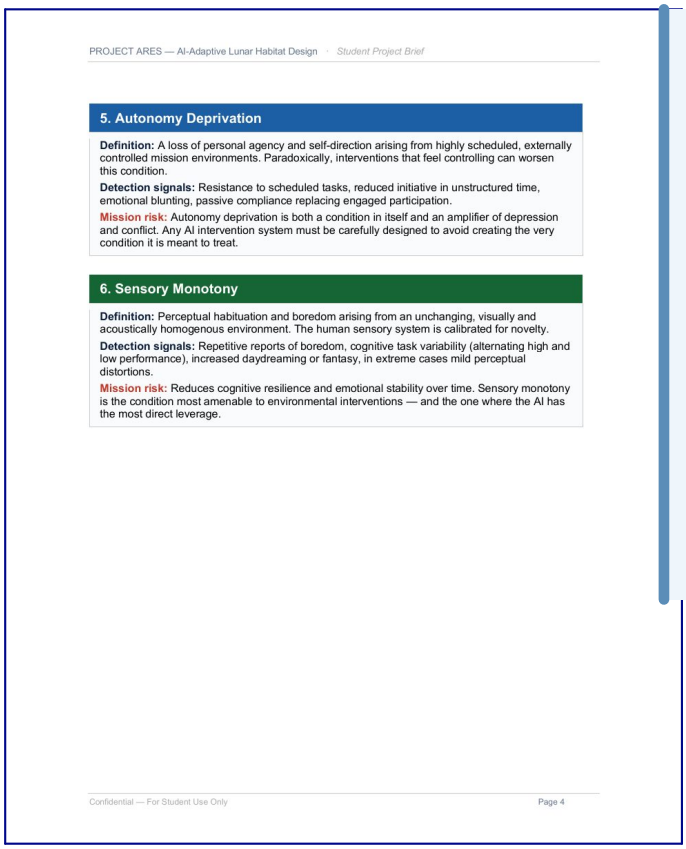
Screenshots:





Realistic game, AI dashboard

Sci-fi game, AI dashboard



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5. Autonomy Deprivation

Definition: A loss of personal agency and self-direction arising from highly scheduled, externally controlled mission environments. Paradoxically, interventions that feel controlling can worsen this condition.

Detection signals: Resistance to scheduled tasks, reduced initiative in unstructured time, emotional blunting, passive compliance replacing engaged participation.

Mission risk: Autonomy deprivation is both a condition in itself and an amplifier of depression and conflict. Any AI intervention system must be carefully designed to avoid creating the very condition it is meant to treat.

6. Sensory Monotony

Definition: Perceptual habituation and boredom arising from an unchanging, visually and acoustically homogenous environment. The human sensory system is calibrated for novelty.

Detection signals: Repetitive reports of boredom, cognitive task variability (alternating high and low performance), increased daydreaming or fantasy, in extreme cases mild perceptual distortions.

Mission risk: Reduces cognitive resilience and emotional stability over time. Sensory monotony is the condition most amenable to environmental interventions — and the one where the AI has the most direct leverage.

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Done | The six conditions (autonomy and sensory monotony)

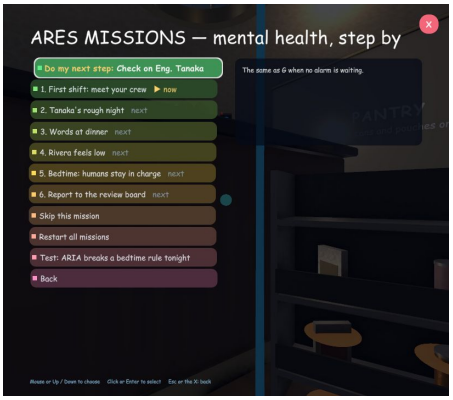
- For autonomy, the AI gives people more choices instead of more control. Options like "let them pick" or "plan your own afternoon" raise it. If someone says no, the AI doesn't keep asking.
- For monotony, the AI slowly shifts the light colour, plays new sounds, rotates scents, and suggests VR nature walks and time in the greenhouse.
- Before the AI does anything, it checks whether the action gives the person more control or less.

How: Press I in the realistic game to get two buttons for each crewmate: check their health, or help them. Help picks the activity that fits whatever that person is struggling with most.

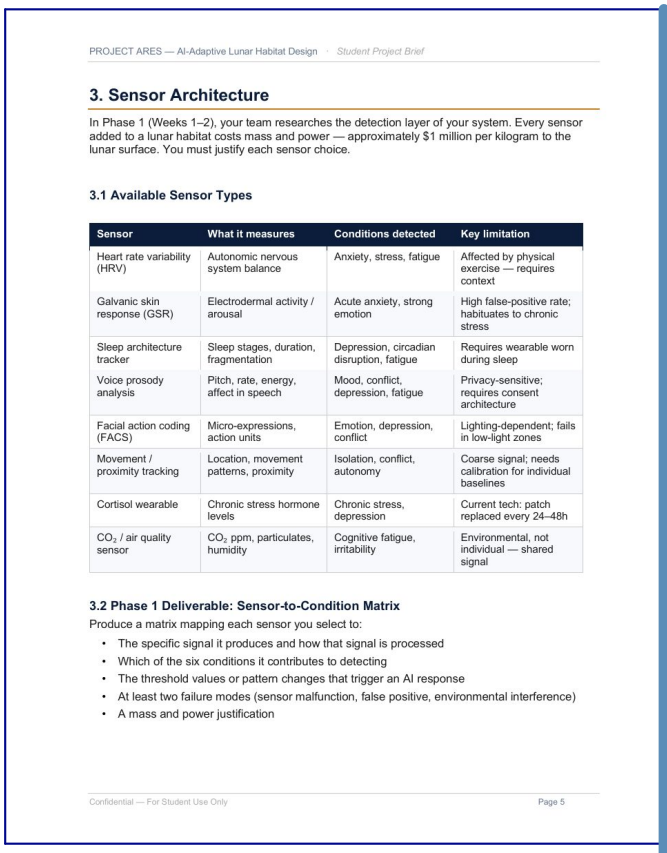
Screenshots:



Realistic game, daily wellness list



Sci-fi game, the six missions



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3. Sensor Architecture

In Phase 1 (Weeks 1–2), your team researches the detection layer of your system. Every sensor added to a lunar habitat costs mass and power — approximately $1 million per kilogram to the lunar surface. You must justify each sensor choice.

3.1 Available Sensor Types

Sensor	What it measures	Conditions detected	Key limitation
Heart rate variability (HRV)	Autonomic nervous system balance	Anxiety, stress, fatigue	Affected by physical exercise — requires context
Galvanic skin response (GSR)	Electrodermal activity / arousal	Acute anxiety, strong emotion	High false-positive rate; habituates to chronic stress
Sleep architecture tracker	Sleep stages, duration, fragmentation	Depression, circadian disruption, fatigue	Requires wearable worn during sleep
Voice prosody analysis	Pitch, rate, energy, affect in speech	Mood, conflict, depression, fatigue	Privacy-sensitive; requires consent architecture
Facial action coding (FACS)	Micro-expressions, action units	Emotion, depression, conflict	Lighting-dependent; fails in low-light zones
Movement / proximity tracking	Location, movement patterns, proximity	Isolation, conflict, autonomy	Coarse signal; needs calibration for individual baselines
Cortisol wearable	Chronic stress hormone levels	Chronic stress, depression	Current tech: patch replaced every 24–48h
CO_2 / air quality sensor	CO_2 ppm, particulates, humidity	Cognitive fatigue, irritability	Environmental, not individual — shared signal

3.2 Phase 1 Deliverable: Sensor-to-Condition Matrix

Produce a matrix mapping each sensor you select to:

- The specific signal it produces and how that signal is processed
- Which of the six conditions it contributes to detecting
- The threshold values or pattern changes that trigger an AI response
- At least two failure modes (sensor malfunction, false positive, environmental interference)
- A mass and power justification

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Done | Sensor architecture and the D1 matrix [Sensor architecture +35]

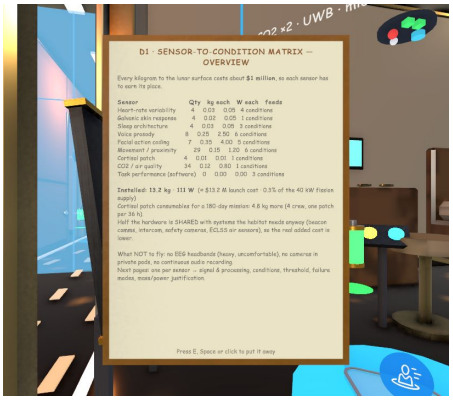
- We picked nine sensors: HRV, skin response (GSR), sleep, voice, facial expressions, location (UWB), a cortisol patch, CO2/air, and task performance, which is just software.
- For each sensor we listed what it measures, which conditions it helps with, the level that triggers a response, two ways it can fail, and its mass and power.
- Altogether it's about 13 kg and 110 W, which is tiny next to the 40 kW reactor. About half the hardware gets shared with other systems like comms and life support.
- The health check-in in the game also shows where each number comes from (wristband, chest beacon, skin patch, room sensors). There's a page that walks through the whole path from the person to the AI.

How: All the sensor numbers live in one place in the code, so the matrix pages and the game always match.

Screenshots:



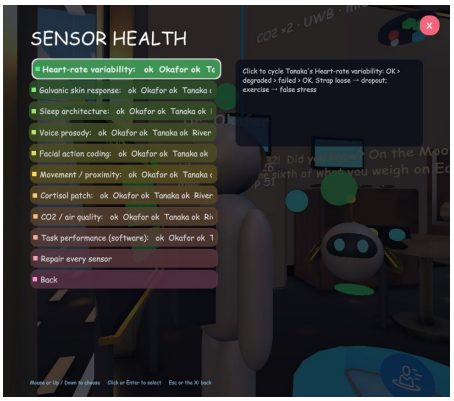
Realistic game, D1 matrix



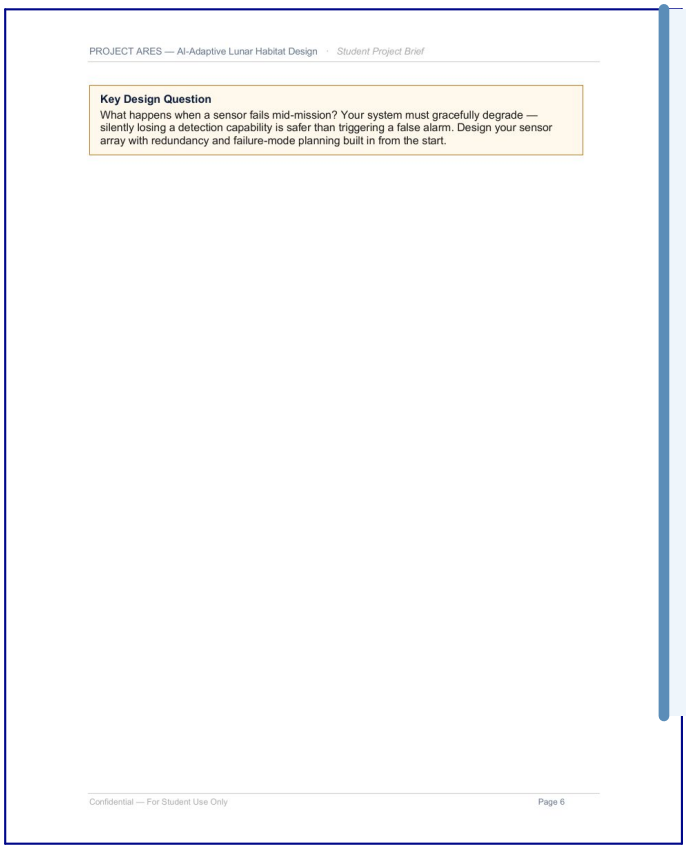
Sci-fi game, D1 matrix



Realistic game, how the data gets to the AI



Sci-fi game, sensor health


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Key Design Question

What happens when a sensor fails mid-mission? Your system must gracefully degrade — silently losing a detection capability is safer than triggering a false alarm. Design your sensor array with redundancy and failure-mode planning built in from the start.

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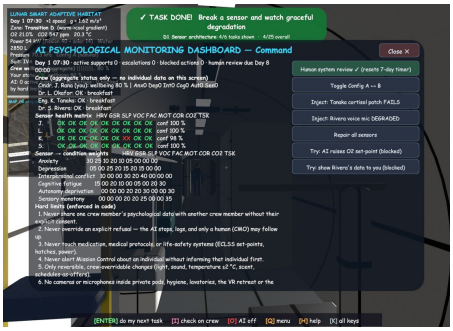
Page 6

Done | What happens if a sensor fails mid-mission [Sensor architecture +35]

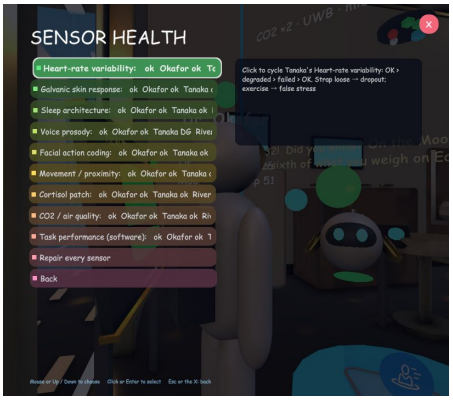
- If a sensor breaks, the AI stops using it and leans more on the others. If it's only giving noisy data, it counts for half.
- If the AI is less than 50% sure, it only does small quiet changes (Tier 1). We decided that missing something is safer than a false alarm.
- Every condition is covered by at least three sensors, so one failure never blinds the system. The life-safety alarms are separate, and the AI can't touch them.
- You can break a sensor yourself in both games and watch the dashboard react.

How: The AI's confidence goes down as sensors drop out, and the game won't let it go past Tier 1 when it's unsure.

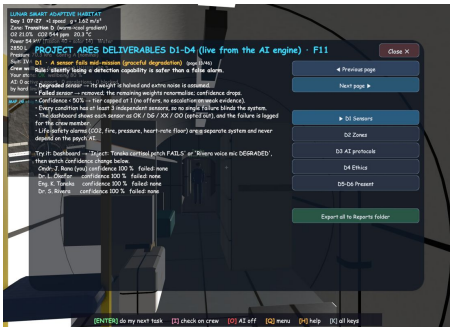
Screenshots:



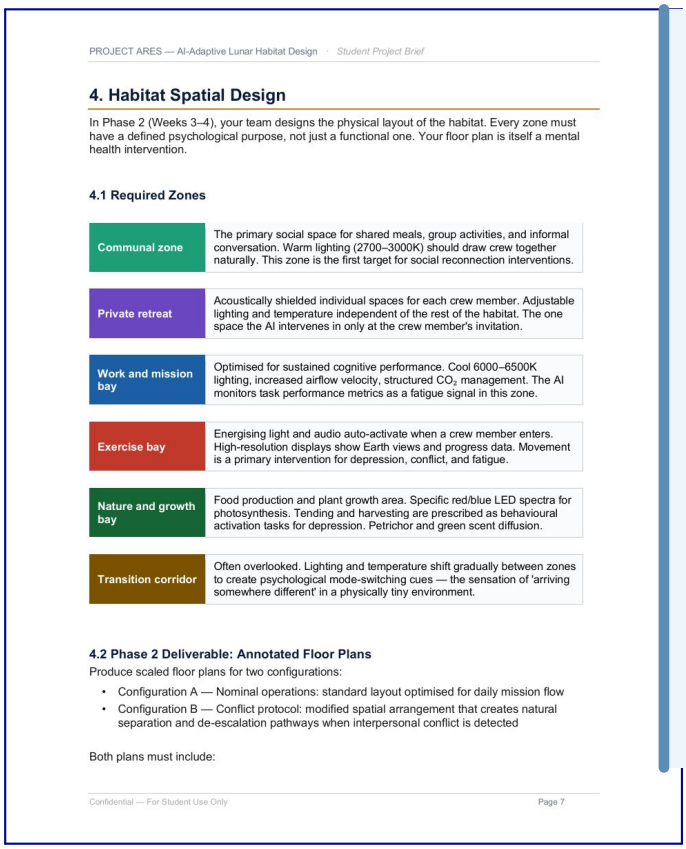
Realistic game, sensor broken, confidence drops



Sci-fi game, Tanaka's heart sensor failed



Realistic game, sensor failure page



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4. Habitat Spatial Design

In Phase 2 (Weeks 3–4), your team designs the physical layout of the habitat. Every zone must have a defined psychological purpose, not just a functional one. Your floor plan is itself a mental health intervention.

4.1 Required Zones

Communal zone	The primary social space for shared meals, group activities, and informal conversation. Warm lighting (2700–3000K) should draw crew together naturally. This zone is the first target for social reconnection interventions.
Private retreat	Acoustically shielded individual spaces for each crew member. Adjustable lighting and temperature independent of the rest of the habitat. The one space the AI intervenes in only at the crew member's invitation.
Work and mission bay	Optimised for sustained cognitive performance. Cool 6000–6500K lighting, increased airflow velocity, structured CO_2 management. The AI monitors task performance metrics as a fatigue signal in this zone.
Exercise bay	Energising light and audio auto-activate when a crew member enters. High-resolution displays show Earth views and progress data. Movement is a primary intervention for depression, conflict, and fatigue.
Nature and growth bay	Food production and plant growth area. Specific red/blue LED spectra for photosynthesis. Tending and harvesting are prescribed as behavioural activation tasks for depression. Petrichor and green scent diffusion.
Transition corridor	Often overlooked. Lighting and temperature shift gradually between zones to create psychological mode-switching cues — the sensation of 'arriving somewhere different' in a physically tiny environment.

4.2 Phase 2 Deliverable: Annotated Floor Plans

Produce scaled floor plans for two configurations:

- Configuration A — Nominal operations: standard layout optimised for daily mission flow
- Configuration B — Conflict protocol: modified spatial arrangement that creates natural separation and de-escalation pathways when interpersonal conflict is detected

Both plans must include:

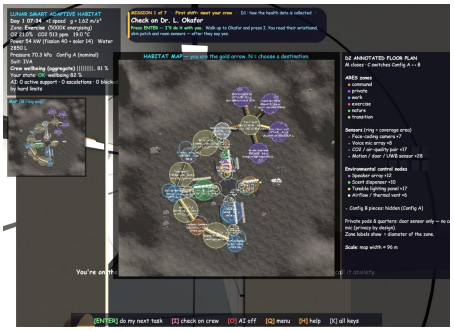
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Done | Habitat spatial design and the six zones [Spatial design +35]

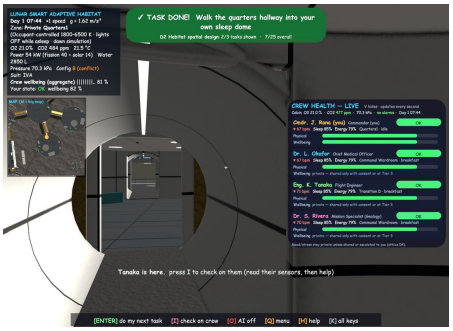
- Both bases have all six zone types, and each one has a purpose. Communal rooms are warm light, private pods are for retreat, the work bay has bright cool light and more airflow, plus there's an exercise bay, a greenhouse and transition hallways.
- Private pods have no camera and no mic, only a door sensor. People set their own light and temperature, and the AI only acts inside if it's asked.
- In the sci-fi game, every room now has its own zone light, a sensor cluster on the ceiling, and 13 hallways where the floor lights fade from one zone's colour to the next. We kept the center dorm where it was.

How: Every room in the game is tagged with its zone type, and the "visit all 6 zones" task uses the same tags.

Screenshots:



Realistic game, map of every zone



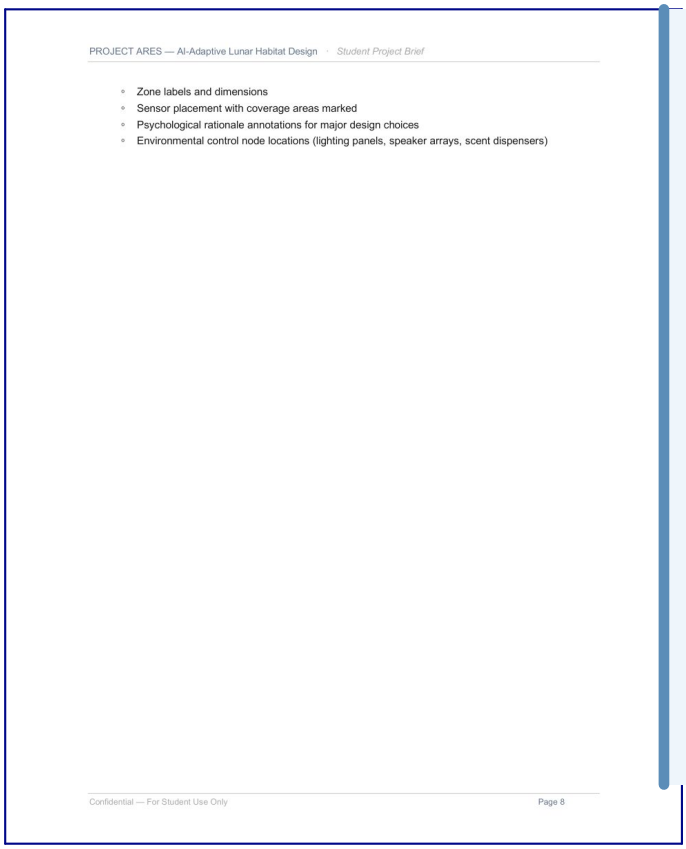
Realistic game, private sleep dome



Sci-fi game, greenhouse



Sci-fi game, exercise bay



Mostly done | Floor plans: Configuration A and B [Spatial design +35]

- Both layouts actually exist in the game world. Config B adds sound screens, a second place to eat and a lit-up second route, so two people who are fighting never have to be together and nobody gets called out.
- Config B only turns on after a person says yes (the crew member accepts it, or the commander switches it).
- The floor plan pages show the zone labels, where the sensors are and what they cover, and where the control panels are.

How: You can switch layouts from the dashboard or by pressing C.

Still need to do: We still need a printed floor plan with real measurements. The in-game map shows the layout but not exact sizes, so we'll do that in Blender or on the 3D model.

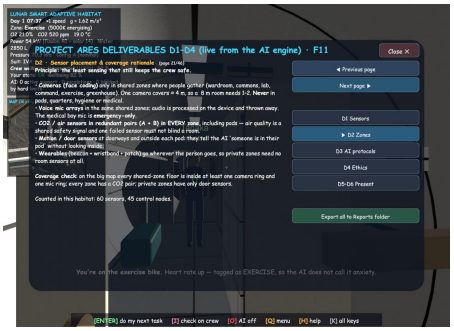
Screenshots:



Realistic game, Config B on



Realistic game, floor plan page



Realistic game, sensor placement



Sci-fi game, Config B screens

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5. AI Response System Design

In Phase 3 (Weeks 5–6), you design the brain of your system — the complete response protocols the AI executes when it detects a psychological state change.

5.1 The Three-Tier Severity Model

Every response protocol must implement a three-tier severity triage:

Tier 1 — Mild / Early
The AI acts silently in the background. No crew notification. Environmental adjustments are subtle and deniable — the crew member may not notice anything has changed. This tier protects crew autonomy and prevents the system from feeling intrusive.

Tier 2 — Moderate
The AI makes more substantial environmental changes and sends a gentle notification to the affected crew member. Phrased as an offer, not an alert: "I've noticed you might benefit from a break — I've adjusted the lounge lighting if you'd like to use it." Options are offered, not imposed.

Tier 3 — Severe
Immediate escalation to the Chief Medical Officer (CMO) and logging for mission control review. The AI continues Tier 2 environmental interventions while the human chain of care activates. The crew member must always be informed that this escalation has occurred.

5.2 Environmental Control Levers

The AI has six domains of environmental control available. For each condition, you must specify which levers are activated, at what intensity, and in what sequence:

Lighting

- Colour temperature (CCT): tunable 1800K–6500K
- Illuminance level: 0–10,000 lux (including clinical bright-light therapy at 10,000 lux)
- Blue-light suppression after 20:00 (mandatory for circadian health)
- Zone-by-zone independent control
- Dynamic hue variation to combat sensory monotony

Audio

- Binaural beat frequency: delta (1–4Hz sleep) through gamma (40Hz peak cognition)

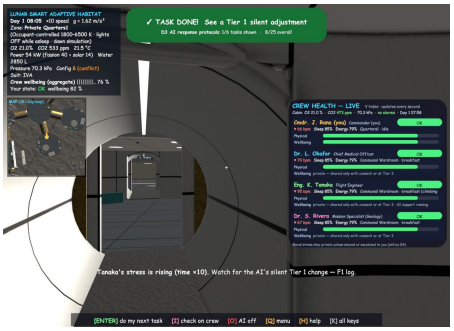
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Done | The three-tier response system [AI protocols +50]

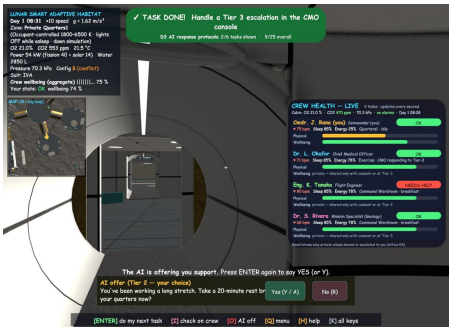
- Tier 1: small quiet changes to the room that can be undone, at most 3 a day per person.
- Tier 2: the AI offers something in plain words, like "want to take a 20 minute break?" If you say no, it stops right away and waits 6 hours.
- Tier 3: the person hears about it first, then the doctor (CMO), who has to respond within 15 minutes. If the CMO is out on EVA, there's a backup (Q6).
- You can trigger and answer each tier yourself in both games.

How: Offers show up as real choices you answer in the game. Tier 3 cases go to the CMO screen.

Screenshots:



Realistic game, Tier 1 change



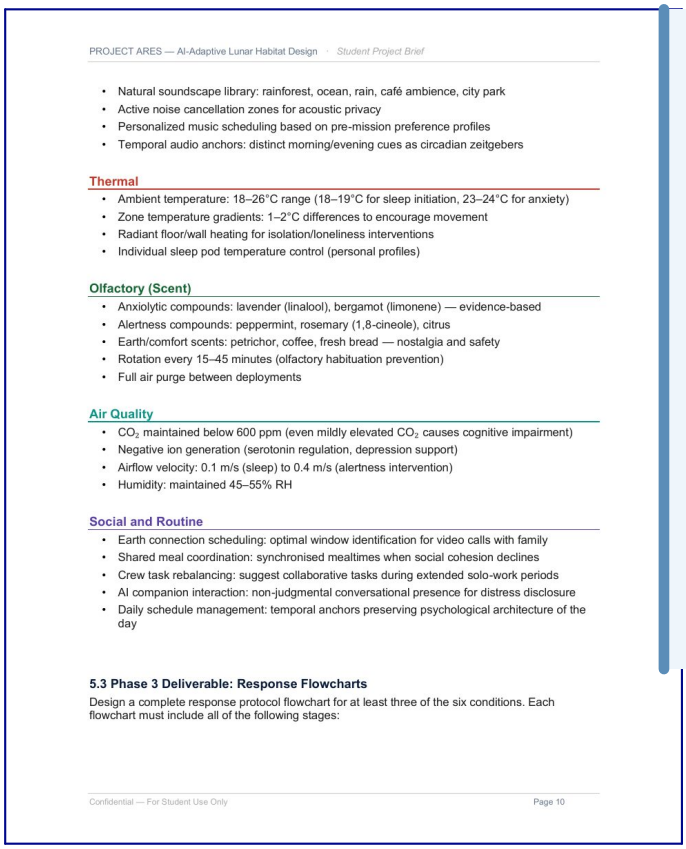
Realistic game, Tier 2 offer



Realistic game, Tier 3 on the CMO screen



Sci-fi game, offer in the crew screen



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- Natural soundscape library: rainforest, ocean, rain, café ambience, city park
- Active noise cancellation zones for acoustic privacy
- Personalized music scheduling based on pre-mission preference profiles
- Temporal audio anchors: distinct morning/evening cues as circadian zeitgebers

Thermal

- Ambient temperature: 18–26°C range (18–19°C for sleep initiation, 23–24°C for anxiety)
- Zone temperature gradients: 1–2°C differences to encourage movement
- Radiant floor/wall heating for isolation/loneliness interventions
- Individual sleep pod temperature control (personal profiles)

Olfactory (Scent)

- Anxiolytic compounds: lavender (linalool), bergamot (limonene) — evidence-based
- Alertness compounds: peppermint, rosemary (1,8-cineole), citrus
- Earth/comfort scents: petrichor, coffee, fresh bread — nostalgia and safety
- Rotation every 15–45 minutes (olfactory habituation prevention)
- Full air purge between deployments

Air Quality

- CO_2 maintained below 600 ppm (even mildly elevated CO_2 causes cognitive impairment)
- Negative ion generation (serotonin regulation, depression support)
- Airflow velocity: 0.1 m/s (sleep) to 0.4 m/s (alertness intervention)
- Humidity: maintained 45–55% RH

Social and Routine

- Earth connection scheduling: optimal window identification for video calls with family
- Shared meal coordination: synchronised mealtimes when social cohesion declines
- Crew task rebalancing: suggest collaborative tasks during extended solo-work periods
- AI companion interaction: non-judgmental conversational presence for distress disclosure
- Daily schedule management: temporal anchors preserving psychological architecture of the day

5.3 Phase 3 Deliverable: Response Flowcharts

Design a complete response protocol flowchart for at least three of the six conditions. Each flowchart must include all of the following stages:

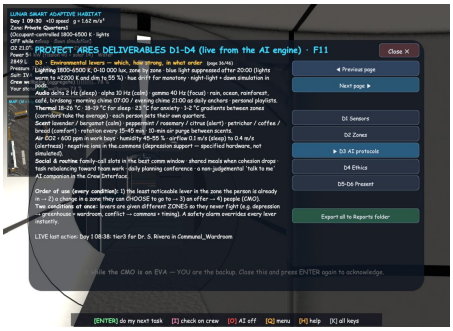
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Done | Environmental controls (light, sound, temperature, scent, air, social) [AI protocols +50]

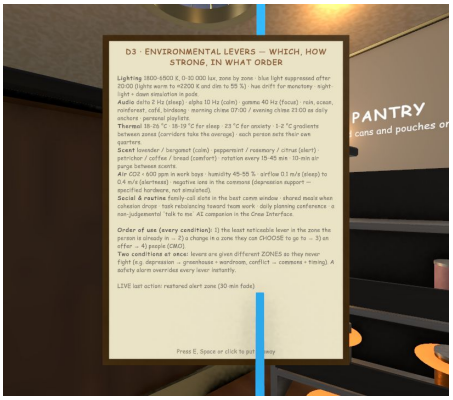
- We used all six types of controls with the numbers from the brief: lights from 1800 to 6500 K, with less blue light after 8pm. There are soundscapes, temperatures from 18 to 26 °C, and scents like lavender, peppermint and "rain". CO2 stays under 600 ppm, and there are social things like shared meals and calls home.
- Each condition has its own order of controls. For anxiety, the AI dims the light to warm amber, plays calm audio, sets the room to 23 °C and adds lavender.

How: Changes fade back to normal over about 30 minutes. A page in the game lists the order for each condition.

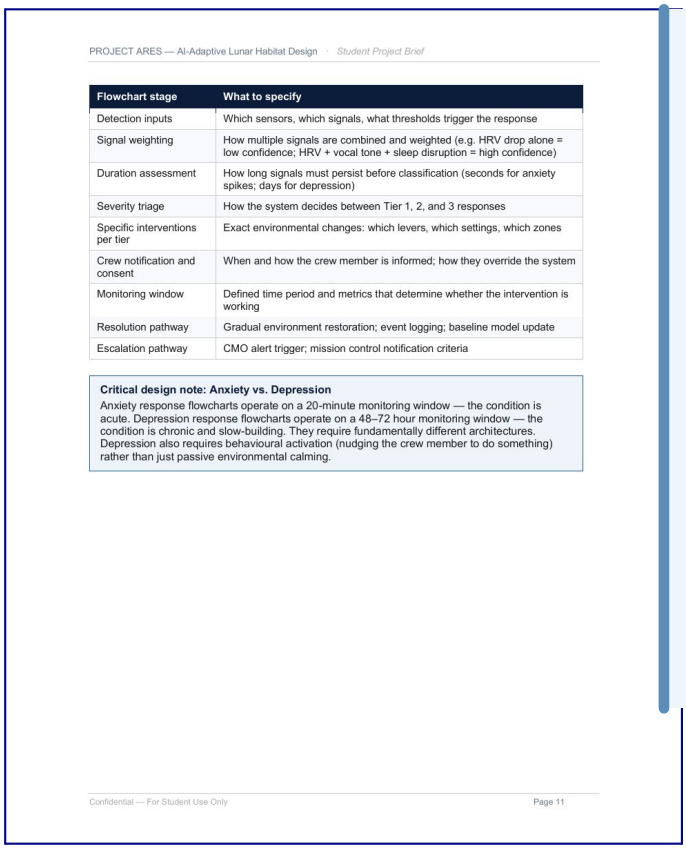
Screenshots:



Realistic game, controls page



Sci-fi game, controls page



PROJECT ARES — AI-Adaptive Lunar Habitat Design · *Student Project Brief*

Flowchart stage	What to specify
Detection inputs	Which sensors, which signals, what thresholds trigger the response
Signal weighting	How multiple signals are combined and weighted (e.g. HRV drop alone = low confidence; HRV + vocal tone + sleep disruption = high confidence)
Duration assessment	How long signals must persist before classification (seconds for anxiety spikes; days for depression)
Severity triage	How the system decides between Tier 1, 2, and 3 responses
Specific interventions per tier	Exact environmental changes: which levers, which settings, which zones
Crew notification and consent	When and how the crew member is informed; how they override the system
Monitoring window	Defined time period and metrics that determine whether the intervention is working
Resolution pathway	Gradual environment restoration; event logging; baseline model update
Escalation pathway	CMO alert trigger; mission control notification criteria

Critical design note: Anxiety vs. Depression

Anxiety response flowcharts operate on a 20-minute monitoring window — the condition is acute. Depression response flowcharts operate on a 48–72 hour monitoring window — the condition is chronic and slow-building. They require fundamentally different architectures. Depression also requires behavioural activation (nudging the crew member to do something) rather than just passive environmental calming.

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Mostly done | D3 response flowcharts [AI protocols +50]

- We wrote protocols for all six conditions, even though the brief only asks for three. Each one covers all nine stages, from detection to escalation.
- Anxiety uses a 20 minute window. Depression uses 48 to 72 hours, and the AI tries to get people doing things (greenhouse, calls home) instead of only calming them down, which is what the brief warns about.
- The game actually runs these steps, so you can watch a condition go through each stage live.

How: The triage page in the game shows the rules for moving a tier up or down.

Still need to do: In the games the flowcharts are text pages. We should also put drawn diagrams on the slides (the anxiety one is already in our final deck).

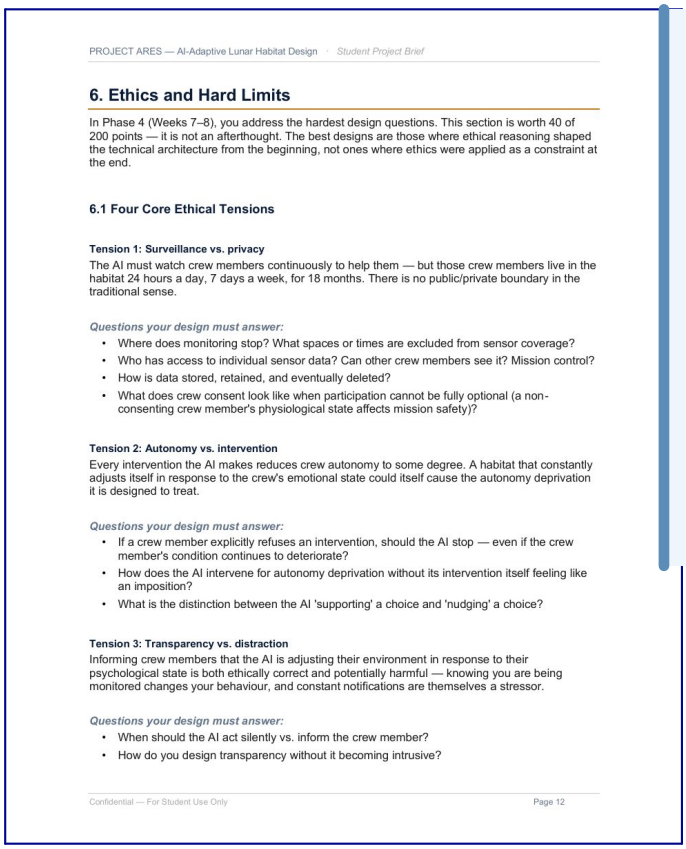
Screenshots:



Realistic game, triage page



Sci-fi game, triage page



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6. Ethics and Hard Limits

In Phase 4 (Weeks 7–8), you address the hardest design questions. This section is worth 40 of 200 points — it is not an afterthought. The best designs are those where ethical reasoning shaped the technical architecture from the beginning, not ones where ethics were applied as a constraint at the end.

6.1 Four Core Ethical Tensions

Tension 1: Surveillance vs. privacy

The AI must watch crew members continuously to help them — but those crew members live in the habitat 24 hours a day, 7 days a week, for 18 months. There is no public/private boundary in the traditional sense.

Questions your design must answer:

- Where does monitoring stop? What spaces or times are excluded from sensor coverage?
- Who has access to individual sensor data? Can other crew members see it? Mission control?
- How is data stored, retained, and eventually deleted?
- What does crew consent look like when participation cannot be fully optional (a non-consenting crew member's physiological state affects mission safety)?

Tension 2: Autonomy vs. intervention

Every intervention the AI makes reduces crew autonomy to some degree. A habitat that constantly adjusts itself in response to the crew's emotional state could itself cause the autonomy deprivation it is designed to treat.

Questions your design must answer:

- If a crew member explicitly refuses an intervention, should the AI stop — even if the crew member's condition continues to deteriorate?
- How does the AI intervene for autonomy deprivation without its intervention itself feeling like an imposition?
- What is the distinction between the AI 'supporting' a choice and 'nudging' a choice?

Tension 3: Transparency vs. distraction

Informing crew members that the AI is adjusting their environment in response to their psychological state is both ethically correct and potentially harmful — knowing you are being monitored changes your behaviour, and constant notifications are themselves a stressor.

Questions your design must answer:

- When should the AI act silently vs. inform the crew member?
- How do you design transparency without it becoming intrusive?

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Done | Ethics: privacy and autonomy [Ethics +40]

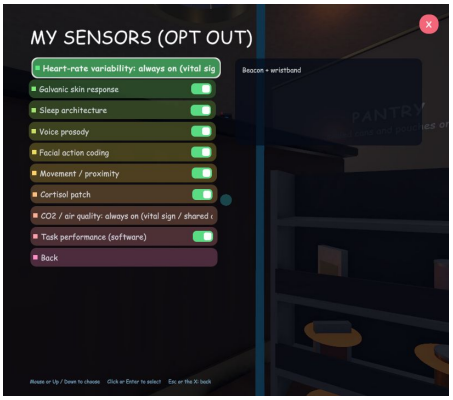
- Monitoring stops at the pod door. There are no cameras or mics in pods, bathrooms or medical, and raw audio and video never leave the room, only numbers do.
- Anyone can turn off any of the mental health sensors. Heart rate and CO2 stay on because they're about safety.
- If someone refuses help, the AI stops and writes it in the log. After that, only a human can follow up.

How: Each person has their own on/off switches in the crew screen. The AI is blocked for 6 hours after someone says no.

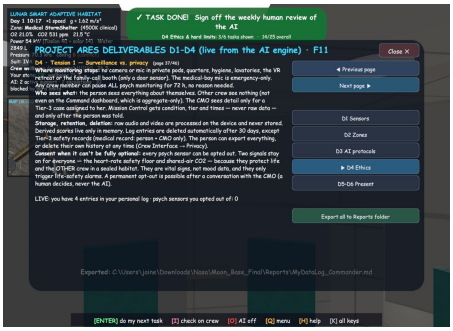
Screenshots:



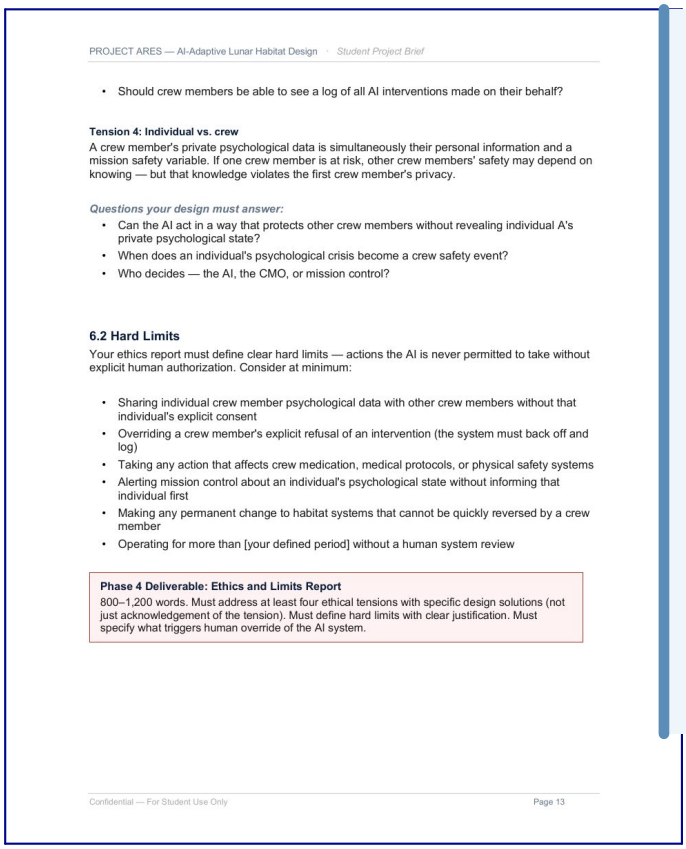
Realistic game, privacy settings



Sci-fi game, turning a sensor off



Realistic game, privacy page



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- Should crew members be able to see a log of all AI interventions made on their behalf?

Tension 4: Individual vs. crew

A crew member's private psychological data is simultaneously their personal information and a mission safety variable. If one crew member is at risk, other crew members' safety may depend on knowing — but that knowledge violates the first crew member's privacy.

Questions your design must answer:

- Can the AI act in a way that protects other crew members without revealing individual A's private psychological state?
- When does an individual's psychological crisis become a crew safety event?
- Who decides — the AI, the CMO, or mission control?

6.2 Hard Limits

Your ethics report must define clear hard limits — actions the AI is never permitted to take without explicit human authorization. Consider at minimum:

- Sharing individual crew member psychological data with other crew members without that individual's explicit consent
- Overriding a crew member's explicit refusal of an intervention (the system must back off and log)
- Taking any action that affects crew medication, medical protocols, or physical safety systems
- Alerting mission control about an individual's psychological state without informing that individual first
- Making any permanent change to habitat systems that cannot be quickly reversed by a crew member
- Operating for more than [your defined period] without a human system review

Phase 4 Deliverable: Ethics and Limits Report

800–1,200 words. Must address at least four ethical tensions with specific design solutions (not just acknowledgement of the tension). Must define hard limits with clear justification. Must specify what triggers human override of the AI system.

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Done | More ethics and the hard limits [Ethics +40]

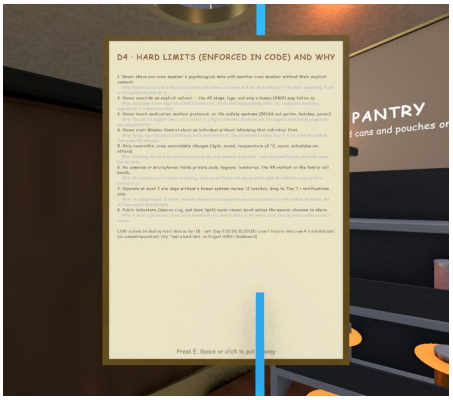
- Tier 1 changes are quiet, but they always show up in the person's own log. Anyone can also ask to be told about every change.
- The AI only changes the room and the schedule. It never tells other people someone's private info. A crisis only becomes a crew thing at Tier 3, and the doctor decides that.
- Every hard limit from the brief is built into the code, and you can test them in the game. That covers no sharing data, no overriding a no, no touching meds or safety systems, no telling mission control before the person knows, only changes that can be undone, and a human review every week.

How: When the AI tries something that isn't allowed, it gets blocked and counted on the dashboard. "Export my data" saves everything it recorded about you.

Screenshots:



Realistic game, hard limits



Sci-fi game, hard limits



Realistic game, export my data



Sci-fi game, exported data

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7. Key Design Constraints

These constraints apply throughout all phases. Designs that ignore them will lose marks in the review board Q&A.

Mass and power budget	Every component sent to the Moon costs mass (approximately $1M/kg to the lunar surface) and draws power (every watt is power not available for life support). For each sensor and actuator you add, you must justify it in terms of mass, power draw, and mission-critical value. There are no free additions.
Crew autonomy is non-negotiable	No intervention may permanently override a crew member's explicit refusal. Your consent architecture must be built in from the beginning — not retrofitted. The system must have a clear and simple override mechanism accessible to every crew member at all times.
Minimal surveillance footprint	Raw sensor data must not be accessible to other crew members. The system observes individuals, not for individuals. Any crew member must be able to request a complete log of what the AI has recorded about them.
Failure mode planning	What happens when a sensor fails, the AI misclassifies a condition, or two signals directly contradict each other? Your design must gracefully degrade. Silently losing a detection capability is safer than triggering a false alarm. Define your failure modes explicitly.
No intervention that worsens autonomy deprivation	A system that constantly adjusts itself in response to the crew's psychological state can itself cause the autonomy deprivation it is designed to treat. Every intervention must be tested against the question: 'Does this increase or decrease the crew member's sense of agency?'
Human escalation must be fast	The AI is a support layer, not a replacement for human care. The escalation pathway from AI detection to CMO response must be designed for speed and clarity. Define maximum response times for each severity tier.

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Done | Key design constraints

- Every sensor has its weight and power listed (about 13 kg and 110 W total).
- Every crew member can always override the AI: pause it for 1 or 3 days, stop all support, or turn it off completely (O in the realistic game, F12 in the sci-fi game).
- Nobody sees anyone else's raw data, and you can always ask for your full log.
- We added a night watch, since the AI is the only one "awake" at night. If anything goes wrong while the crew sleeps (an alarm, the AI breaking a bedtime rule, an emergency or a broken sensor), the AI turns itself off, undoes its changes and wakes you up. You're in charge until you turn it back on. We added this because AI can be dangerous sometimes.

How: Turning the AI off undoes everything it changed, and only a human can turn it back on.

Screenshots:



Realistic game, AI breaks a rule



Realistic game, AI off switch



Realistic game, night watch stops the AI



Sci-fi game, AI stopped, you're in charge

8. Required Deliverables

#	Deliverable	Due	Description
D1	Sensor-to-condition detection matrix	End of Week 2	A table mapping each sensor you have selected to the conditions it detects, the signal it produces, the detection thresholds you have set, at least two failure modes, and a mass/power justification.
D2	Annotated habitat floor plans (×2 configurations)	End of Week 4	Scaled drawings with zone labels, sensor placement and coverage, environmental control node locations, and psychological rationale annotations. Configuration A (nominal) and Configuration B (conflict protocol).
D3	AI response flowcharts (minimum 3 of 6 conditions)	End of Week 6	Complete detection → triage → intervention → monitoring → escalation pathway for each chosen condition. All nine flowchart stages (see Section 5.3) must be addressed.
D4	Ethics and limits report	End of Week 7	800–1,200 words. Must address at least four ethical tensions with specific design solutions. Must define hard limits with clear justification. Must specify human override procedures.
D5	Agency review board presentation	Week 8	15-minute team presentation with visual aids, followed by 10-minute Q&A from the panel. All team members must contribute to the presentation. Visual aids must be your own design.
D6	Individual reflection journal (×3 entries)	Weeks 2, 5, and 8	Personal reflections on design decisions made, team process, ethical questions encountered, and evolving thinking. Approximately 300–400 words per entry. Graded individually, not as a team.

8.1 Bonus Challenges (+20 Points Available)

Bonus A: Tabletop simulation (+10 points)

Run a structured tabletop simulation of a 72-hour mission window in which two scripted psychological events occur simultaneously. Document your system's response, where it succeeded, and where it failed. Identify what you would redesign as a result. Submit as a written simulation report (600–900 words) with supporting diagrams.

Bonus B: Crew interface mockup (+10 points)

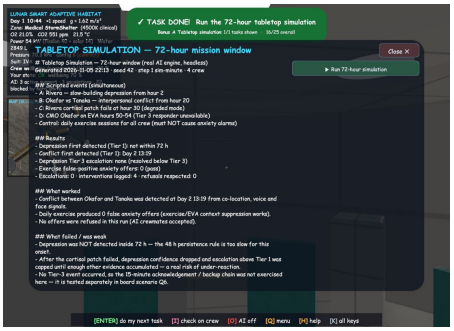
Design a screen-based interface through which crew members can: view their own wellness data, see what the AI has done on their behalf, override any active intervention, and communicate preferences to the system. Must include privacy controls, a complete override mechanism, and an opt-out pathway.

Done | Deliverables D1 to D6 and the bonus challenges [Bonus A +10 · Bonus B +10]

- D1 matrix (page 5), D2 floor plans (pages 7 and 8), D3 flowcharts (page 11) and the D4 ethics report are all in both games, and the D4 pages can be exported as one report.
- D5: the presentation order is built into the game, and all six board questions run live (page 21). D6: the journal prompts are in the game, but we have to write the journals ourselves.
- Bonus A: the 72 hour tabletop simulation runs in the game and saves a report. Bonus B: the crew screen shows wellness, what the AI did for you, override, privacy, opt-out and preferences.

How: Each deliverable is also a task on the checklist in the realistic game (press J).

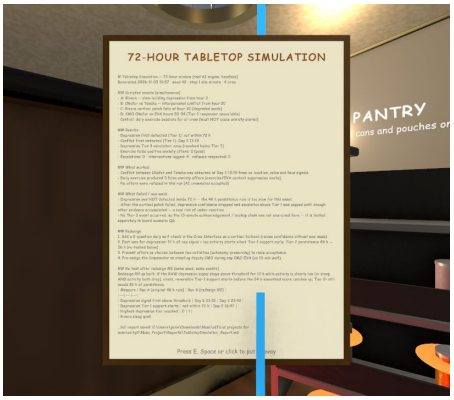
Screenshots:



Realistic game, Bonus A tabletop



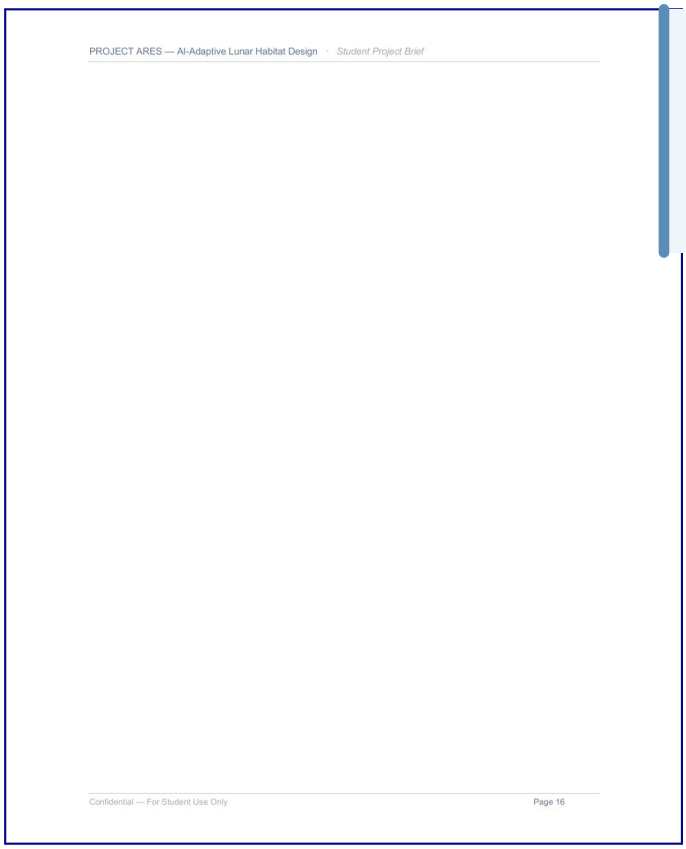
Realistic game, Bonus B crew screen



Sci-fi game, tabletop report



Sci-fi game, crew screen



Done | Bonus challenges (continued) [Bonus A +10 · Bonus B +10]

- Same as page 15. Both bonus challenges are done in both games.

Screenshots:



Realistic game, presentation order page

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9. Grading Rubric — 200 Points

Sensor architecture and detection logic	**35 pts**	*Graded as team*

- Accuracy and completeness of sensor-to-condition mapping
- Realistic and justified detection thresholds
- Thoughtful handling of false positives and signal ambiguity
- Mass and power justification for each sensor
- At least two failure modes addressed per sensor

Habitat spatial design	**35 pts**	*Graded as team*

- Clear psychological purpose for each of the six zones
- Logical sensor placement with coverage rationale
- Credible comparison between two configurations
- Annotations demonstrate evidence-based design decisions

AI response protocol design	**50 pts**	*Graded as team*

- Completeness of flowcharts — all nine stages present
- Evidence basis for specific interventions chosen
- Sound severity triage logic with defined thresholds
- Consent and crew notification architecture clearly specified
- Monitoring window metrics defined and measurable
- Escalation pathway to CMO is fast and clear

Ethics and limits analysis	**40 pts**	*Graded as team*

- Depth and genuine engagement with at least four ethical tensions
- Quality of proposed design solutions (not just acknowledgement)
- Clarity, defensibility, and completeness of hard limits
- Evidence that ethical reasoning shaped the technical design

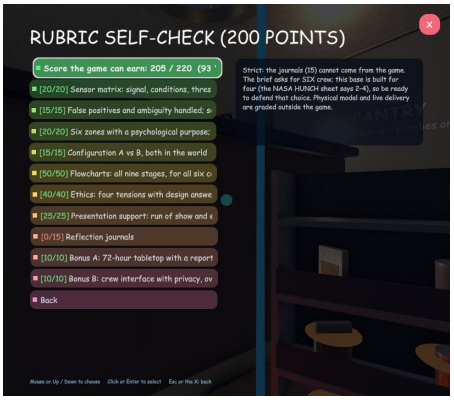
Presentation and communication	**25 pts**	*Graded as team*

- Clarity and logical structure of 15-minute presentation
- Quality and originality of visual aids
- Team coordination — all members contribute
- Confidence and accuracy under Q&A
- Ability to defend specific design choices with evidence

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Our grade | Grading rubric and our self-grade

Screenshots:

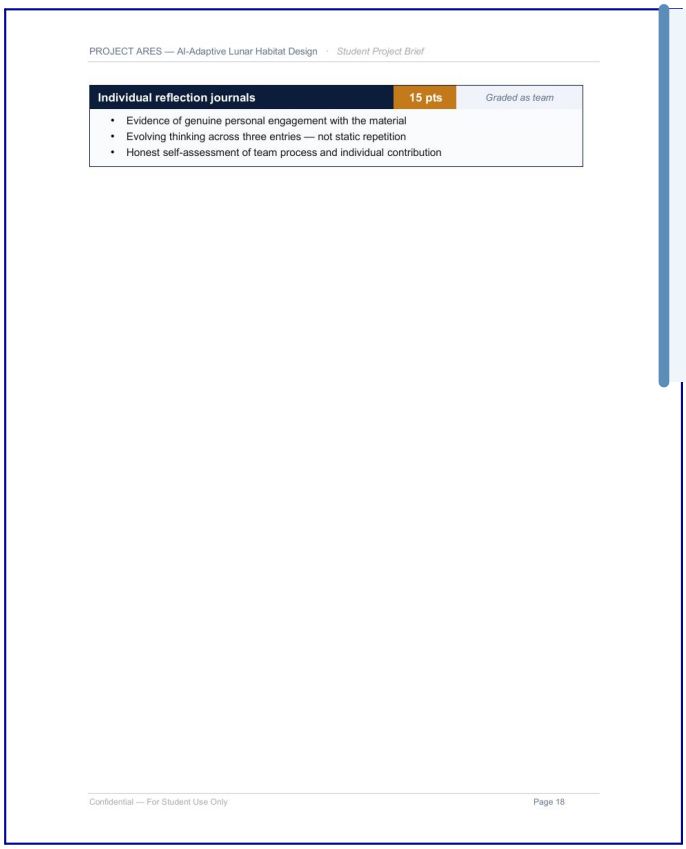


Sci-fi game, rubric check page

Self-grade, line by line (strict):

Category	Points	Ours	Why
Sensor architecture and detection logic	+35	**33/35**	9 sensors with thresholds, failure modes, weight and power, and backup when one breaks. Lost 2 because our thresholds are design guesses, not tested data.
Habitat spatial design	+35	**30/35**	6 zones in both bases, sensor placement, Config A and B. Lost 5 because there's no floor plan with measurements in the game and we have 4 crew instead of 6.
AI response protocol design	+50	**46/50**	All 6 conditions with all 9 stages, running live, with tiers, consent and doctor escalation. Lost 4 because the flowcharts are text in the game, not drawn.

Category	Points	Ours	Why
Ethics and limits analysis	+40	**38/40**	4 tensions answered, hard limits in the code, override and night watch. Lost 2 because we still need to check the report is 800 to 1,200 words.
Presentation and communication	+25	**22/25**	Presentation order and Q1 to Q6 run in the game. Lost 3 because how we present is judged on the day.
Individual reflection journals	+15	**0/15**	We each have to write these ourselves. The prompts are in the game.
Bonus A: tabletop simulation	+10	**10/10**	72 hour run with two events at once, a broken sensor and the doctor on EVA, plus a saved report.
Bonus B: crew interface mockup	+10	**10/10**	Wellness, what the AI did, override, privacy, opt-out and preferences.
Total (200 + 20 bonus)	**220**	**189/220**	**About 86% overall, or about 92% if you leave out the journals we still have to write.**



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Individual reflection journals	15 pts	*Graded as team*

- Evidence of genuine personal engagement with the material
- Evolving thinking across three entries — not static repetition
- Honest self-assessment of team process and individual contribution

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Our grade | Reflection journals

- The journals are worth 15 points and they're about our own thinking, so the game can't do them for us. The prompts are in the game on the D6 page.
- Ideas for what to write about: why we added the night watch, and why heart rate and CO2 can't be turned off.

Still need to do: Each of us still has to write three 300 to 400 word entries (weeks 2, 5 and 8).

Screenshots:



Realistic game, journal prompts

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10. Eight-Week Timeline

Weeks 1–2
Phase 1: Research and Sensor Architecture

- Review psychological literature on isolated and confined environments (ICE): Antarctic wintering-over data, ISS crew reports, submarine studies
- Research each sensor type: how it works, what it measures, what it costs in mass and power
- Build your sensor-to-condition detection matrix
- Identify at least two failure modes per sensor
- Reflection journal entry 1 due end of Week 2

Deliverable: D1: Sensor-to-condition matrix

Weeks 3–4
Phase 2: Habitat Spatial Design

- Design the physical layout of your habitat module — floor plan, zone functions, dimensions
- Determine sensor placement in each zone with coverage area calculations
- Create Configuration A (nominal) and Configuration B (conflict protocol)
- Write psychological rationale annotations for every major design decision
- Check each design decision against your key constraints list

Deliverable: D2: Annotated floor plans (×2 configurations)

Weeks 5–6
Phase 3: AI Response System Design

- Design complete response protocols for all six conditions
- Produce detailed flowcharts for your chosen three conditions (minimum)
- Design your consent and crew notification architecture
- Define monitoring window metrics for each condition
- Design your escalation pathway from AI detection to CMO to mission control
- Reflection journal entry 2 due end of Week 5

Deliverable: D3: AI response flowcharts (3+ conditions)

Weeks 7–8
Phase 4: Ethics, Hard Limits, and Presentation

- Write your ethics and limits report — treat this as a genuine design document, not an afterthought

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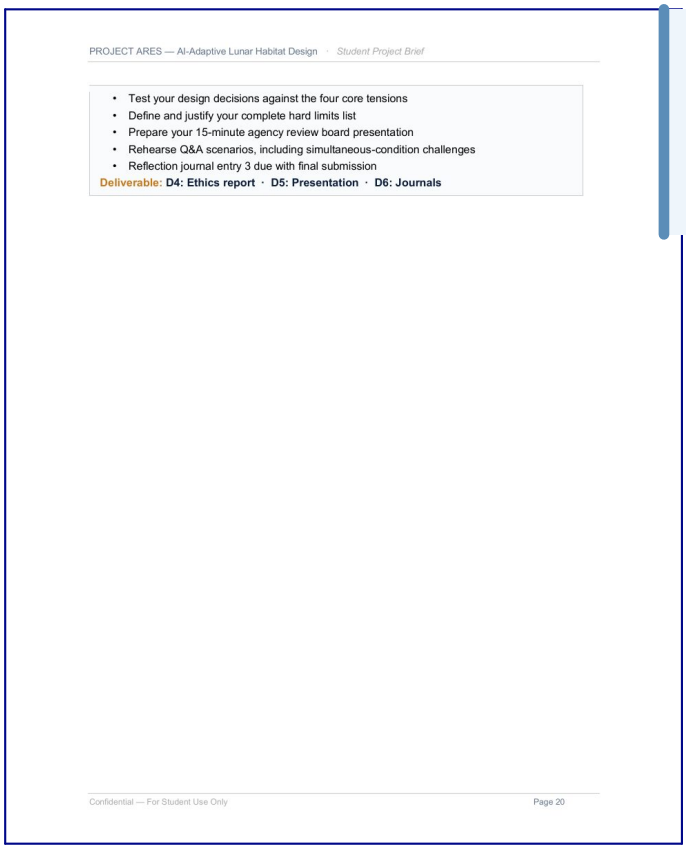
Done | Timeline (weeks 1 to 4)

- Everything from Phase 1 (sensors) and Phase 2 (layout) is in the games and slides. Our research doc covers places like Antarctica, the ISS, Mars-500 and HI-SEAS.

Screenshots:



Realistic game, help screen


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- Test your design decisions against the four core tensions
- Define and justify your complete hard limits list
- Prepare your 15-minute agency review board presentation
- Rehearse Q&A scenarios, including simultaneous-condition challenges
- Reflection journal entry 3 due with final submission

Deliverable: D4: Ethics report · D5: Presentation · D6: Journals

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Done | Timeline (weeks 5 to 8)

- Phase 3 (protocols) and Phase 4 (ethics, presentation and Q&A practice) can all be played, and we can practice the board questions inside the game.

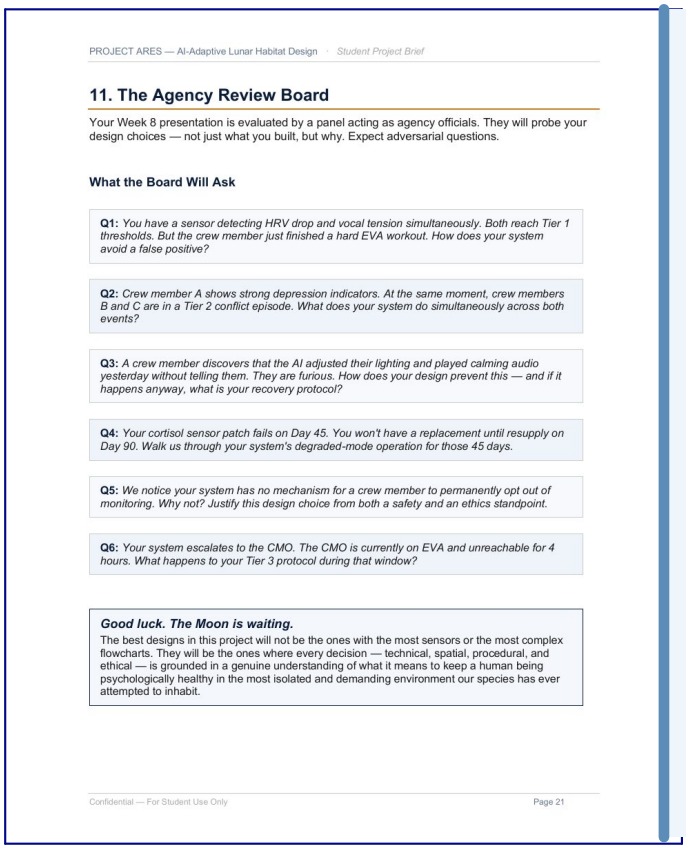
Screenshots:



Realistic game, menu



Realistic game, daily banner



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11. The Agency Review Board

Your Week 8 presentation is evaluated by a panel acting as agency officials. They will probe your design choices — not just what you built, but why. Expect adversarial questions.

What the Board Will Ask

Q1: *You have a sensor detecting HRV drop and vocal tension simultaneously. Both reach Tier 1 thresholds. But the crew member just finished a hard EVA workout. How does your system avoid a false positive?*

Q2: *Crew member A shows strong depression indicators. At the same moment, crew members B and C are in a Tier 2 conflict episode. What does your system do simultaneously across both events?*

Q3: *A crew member discovers that the AI adjusted their lighting and played calming audio yesterday without telling them. They are furious. How does your design prevent this — and if it happens anyway, what is your recovery protocol?*

Q4: *Your cortisol sensor patch fails on Day 45. You won't have a replacement until resupply on Day 90. Walk us through your system's degraded-mode operation for those 45 days.*

Q5: *We notice your system has no mechanism for a crew member to permanently opt out of monitoring. Why not? Justify this design choice from both a safety and an ethics standpoint.*

Q6: *Your system escalates to the CMO. The CMO is currently on EVA and unreachable for 4 hours. What happens to your Tier 3 protocol during that window?*

Good luck. The Moon is waiting.

The best designs in this project will not be the ones with the most sensors or the most complex flowcharts. They will be the ones where every decision — technical, spatial, procedural, and ethical — is grounded in a genuine understanding of what it means to keep a human being psychologically healthy in the most isolated and demanding environment our species has ever attempted to inhabit.

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Done | Agency Review Board questions Q1 to Q6 [Presentation +25]

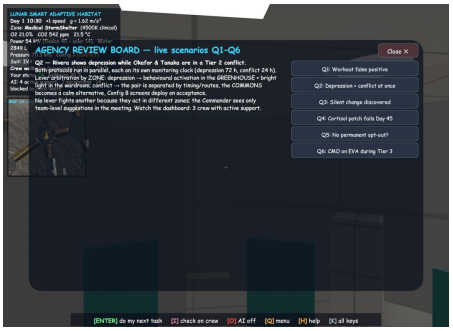
- We answered all six questions, and each one can be run live in both games. The game sets up the situation and then explains what our design does.
- Q1: a workout looks like stress, so the AI knows you're exercising and doesn't panic. Q2: depression and conflict at the same time, handled with separate zones and timers. Q3: someone finds a quiet change they didn't know about, so they can see the log, the AI apologizes, and they can turn Tier 1 off for themselves. Q4: the cortisol patch fails on day 45, so the AI runs in backup mode with a weekly doctor check-in. Q5: why you can't opt out forever, because heart rate is the safety floor and everything else can be paused. Q6: the doctor is out on EVA during a Tier 3, so the commander is the backup, then mission control.

How: You can also press ENTER six times in the realistic game to run all six questions one after another.

Screenshots:



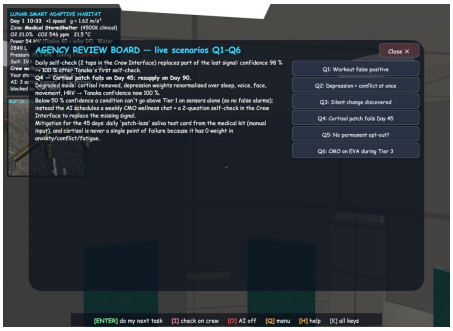
Realistic game, Q1



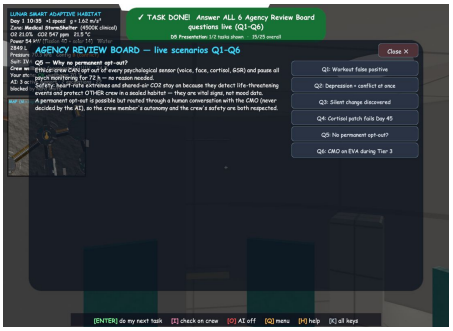
Realistic game, Q2



Realistic game, Q3



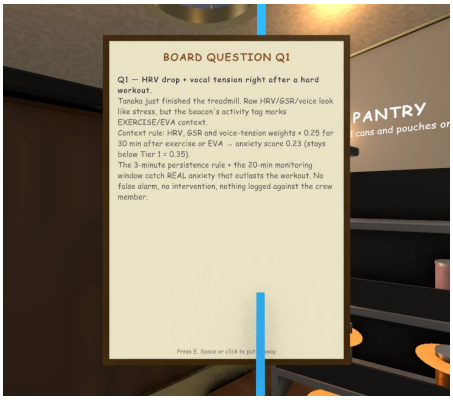
Realistic game, Q4



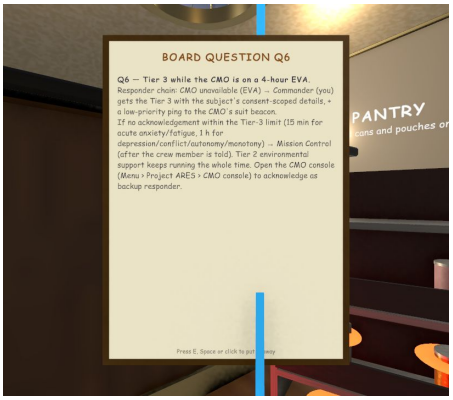
Realistic game, Q5



Realistic game, Q6



Sci-fi game, Q1



Sci-fi game, Q6